

HUALIEN, Taiwan

WMO: 466990

Lat: 23.9751N

Lon: 121.6132E

Elev: 52

StdP: 14.67

Time Zone: 8.00 (E08)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	55.0	56.8	44.8	44.0	59.1	47.0	47.7	60.0	20.9	65.9	18.8	65.5	5.5	240	0.283

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	90.8	81.3	89.5	80.5	88.5	80.2	82.4	89.1	81.6	88.2	80.9	87.4	7.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.4	158.1	87.9	79.6	153.9	87.1	78.9	150.3	86.3	46.3	89.2	45.4	88.5	44.7	87.7	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.9	16.3	13.9	DB	50.9	94.2	1.7	2.9	49.7	96.3	48.7	97.9	47.8	99.6	46.5	101.6	
			WB	47.6	84.2	2.5	1.3	45.9	85.1	44.4	85.9	43.0	86.6	41.2	87.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.5	65.0	64.9	68.8	73.2	77.3	80.8	83.3	82.9	79.8	76.1	72.7	68.4
	DBStd	7.24	4.06	4.45	4.81	4.71	3.53	2.85	1.96	1.95	2.53	2.87	3.26	3.76
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	147	53	52	21	4	0	0	0	0	0	0	2	15
	CDD50	8938	466	417	582	697	845	923	1031	1021	894	810	681	571
	CDD65	3609	53	49	138	251	380	473	566	556	444	345	233	121
	CDH74	30193	42	71	414	1351	2842	4977	6911	6563	4100	1995	772	155
	CDH80	8903	0	0	15	175	587	1584	2733	2457	1019	275	55	3
(14) Wind	WSAvg	5.4	5.7	5.7	5.5	5.3	4.7	5.0	5.2	5.2	5.4	5.7	5.6	6.0
(15) Precipitation	PrecAvg	83.8	2.6	4.0	3.5	4.4	8.0	9.0	7.0	9.3	13.9	14.1	5.2	2.1
	PrecMax	140.1	7.9	13.3	11.4	14.5	23.6	27.5	31.1	26.7	38.0	66.9	16.9	5.6
	PrecMin	38.9	0.5	0.9	0.9	1.0	1.2	1.0	0.6	0.1	1.3	0.5	0.2	0.6
	PrecStd	21.9	1.4	3.0	2.0	3.1	4.8	6.2	6.4	7.4	9.4	14.8	4.4	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.3	78.5	81.7	86.1	87.8	90.3	93.1	92.4	89.7	86.4	83.3	80.2
		MCWB	69.3	71.4	74.3	77.0	79.5	80.3	82.9	81.8	80.1	78.6	75.3	73.3
	2%	DB	75.2	76.3	79.9	83.8	86.3	89.1	91.3	90.5	88.0	84.6	81.7	77.4
		MCWB	68.1	69.7	72.9	76.1	78.6	80.3	81.7	81.2	79.3	76.8	73.9	70.2
	5%	DB	73.2	74.3	78.2	82.1	85.0	88.0	89.8	89.2	86.5	83.1	80.0	75.6
		MCWB	66.6	68.0	72.0	75.3	77.6	80.0	80.6	80.6	78.5	75.5	72.4	68.8
	10%	DB	71.3	72.0	76.4	80.4	83.3	86.8	88.5	88.1	85.0	81.4	78.2	73.9
		MCWB	65.5	66.3	70.8	74.4	76.6	79.3	80.2	80.2	77.5	74.2	71.1	67.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.6	72.8	75.5	78.8	80.8	81.9	83.8	83.3	81.5	80.1	77.1	73.8
		MCDB	76.0	77.2	80.4	83.8	86.9	88.3	91.6	90.3	88.1	85.2	81.4	78.8
	2%	WB	68.8	70.6	73.9	76.9	79.1	81.3	82.6	82.2	80.3	78.3	75.4	71.8
		MCDB	73.8	75.3	78.8	82.6	84.9	87.7	89.6	88.9	86.4	83.0	80.5	76.0
	5%	WB	67.5	68.8	72.5	75.6	78.2	80.5	81.8	81.4	79.3	76.8	73.7	70.1
		MCDB	72.2	73.1	77.4	81.2	83.7	87.0	88.5	88.0	85.0	81.3	78.6	74.4
	10%	WB	66.2	67.3	71.2	74.7	77.1	79.6	80.9	80.7	78.4	75.5	72.3	68.4
		MCDB	70.8	71.3	75.9	80.1	82.5	85.8	87.5	87.1	83.7	79.6	77.0	73.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	8.3	8.8	9.1	9.4	9.6	10.5	10.3	9.5	8.7	9.1	8.3
		MCDBR	12.0	12.1	11.8	11.8	11.5	11.6	12.2	11.9	11.6	11.3	11.7	11.1
	5% WB	MCWBR	7.6	8.4	7.5	8.1	7.8	7.9	7.9	7.3	7.1	6.6	8.0	7.1
		MCWBR	11.0	11.2	11.0	10.9	10.7	11.0	11.3	10.9	10.4	10.3	10.8	9.5
(40) Clear-Sky Solar Irradiance	taub		0.544	0.578	0.700	0.706	0.614	0.551	0.528	0.563	0.562	0.546	0.534	0.549
		taud	1.894	1.853	1.618	1.658	1.883	2.054	2.124	2.009	1.995	1.990	1.996	1.908
	Ebn at Noon		223	226	206	209	229	242	248	239	236	232	225	215
		Edh at Noon	56	62	83	81	64	54	50	56	56	54	50	54
(44) All-Sky Solar Radiation	RadAvg		754	880	1000	1117	1310	1560	1817	1633	1431	1190	875	747
	RadStd		108	142	145	176	215	183	115	161	152	126	93	115

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	+0.49	N/A	N/A	N/A	N/A	N/A	+146

Nomenclature: See separate page